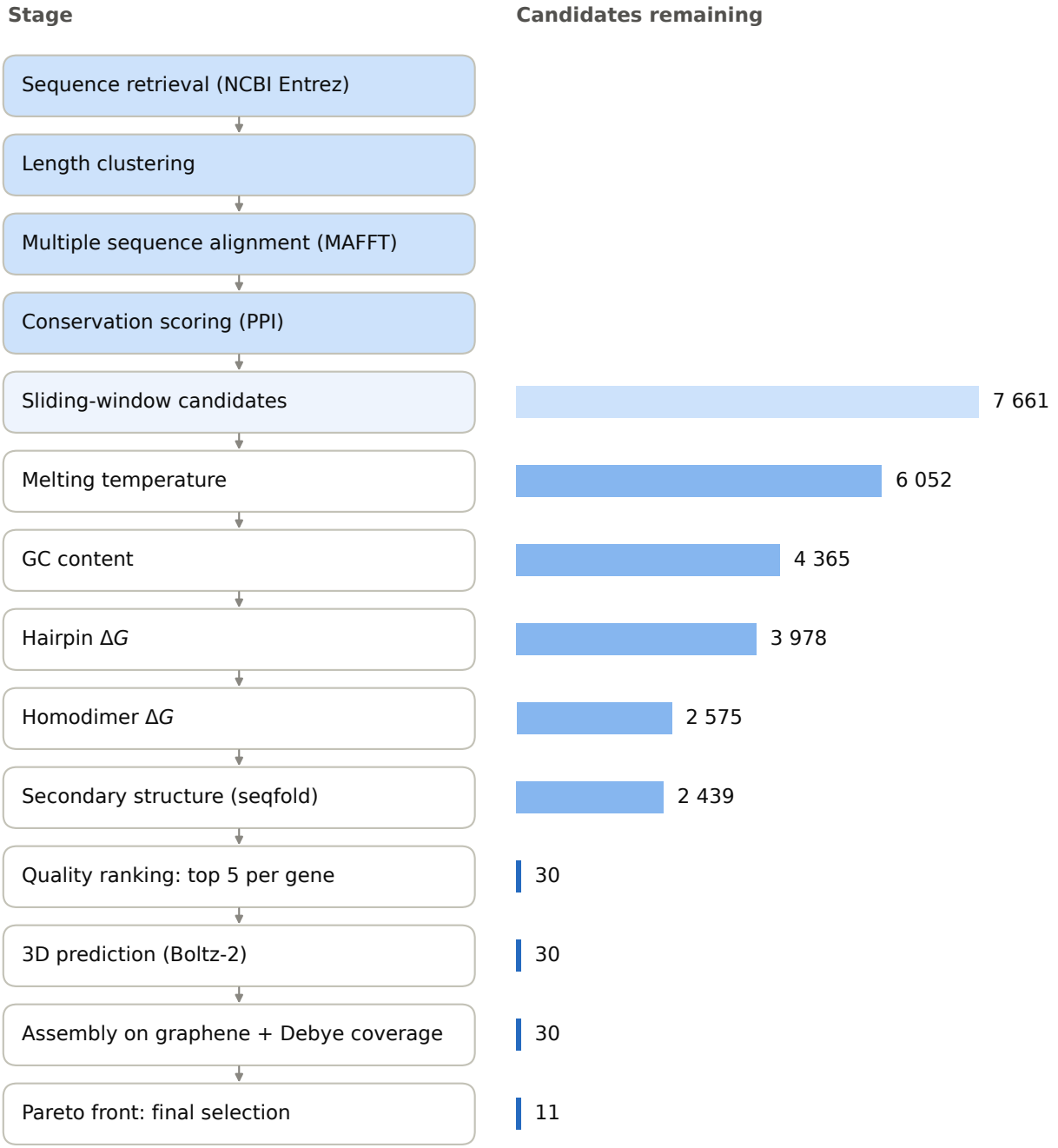
Stage
Candidates remaining
Sequence retrieval (NCBI Entrez)
Length clustering
Multiple sequence alignment (MAFFT)
Conservation scoring (PPI)
Sliding-window candidates
7 661
Melting temperature
6 052
GC content
4 365
Hairpin ΔG
3 978
Homodimer ΔG
2 575
Secondary structure (seqfold)
2 439
Quality ranking: top 5 per gene
30
3D prediction (Boltz-2)
30
Assembly on graphene + Debye coverage
30
Pareto front: final selection
11